XSuM: a generative surrogate of the SBND reconstruction chain trained on the BNB Monte Carlo — technical note

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Abstract

XSuM is a per-event surrogate of the SBND simulation and reconstruction chain: given the true neutrino vertex and the post-FSI final-state particles of an interaction, it samples the record that the cafpyana analysis dataframes hold for that interaction (whether and how many Pandora slices are truth-matched to it, the slice-level block of about fifty reconstructed quantities, and later the set of particle-flow objects), in the same dataframe format as the production. Analyses then run unchanged on the surrogated dataframes. The model is trained on the 2026 gen2 BNB + cosmics central-value Monte Carlo (v10_14_02_03) processed with a selection-free cafpyana configuration written for this purpose. This note documents the data, the inputs and outputs, the baseline and flow-matching models, and their closure on held-out files. It is updated as the production and the models progress.

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1 Purpose and scope

The physics case is forward folding without re-simulation: a cross-section model that was never run through Geant4, Pandora and cafpyana (another generator, a modified FSI model, a new kinematic region) is transported to reconstruction level by the surrogate and compared with data or with the nominal MC at that level. The surrogate does not fix data/MC disagreement; it reproduces the MC detector response, including its flaws, for new inputs.

Scope decisions (2026-10-16): XSuM generates the complete reconstruction record, not analysis quantities. Selection flags, BDT scores and derived kinematics such as $|t|$ are recomputed downstream by the analysers (CC coherent pion: S. Oh; charged kaons: T. Wester) from the generated variables. Cosmic-only slices do not depend on the neutrino model and are reused from the MC. Generator-internal labels (interaction mode, CC/NC, W, Q^2, E_ν, signal flags) are never inputs; they are kept in a separate label table for holdout tests.

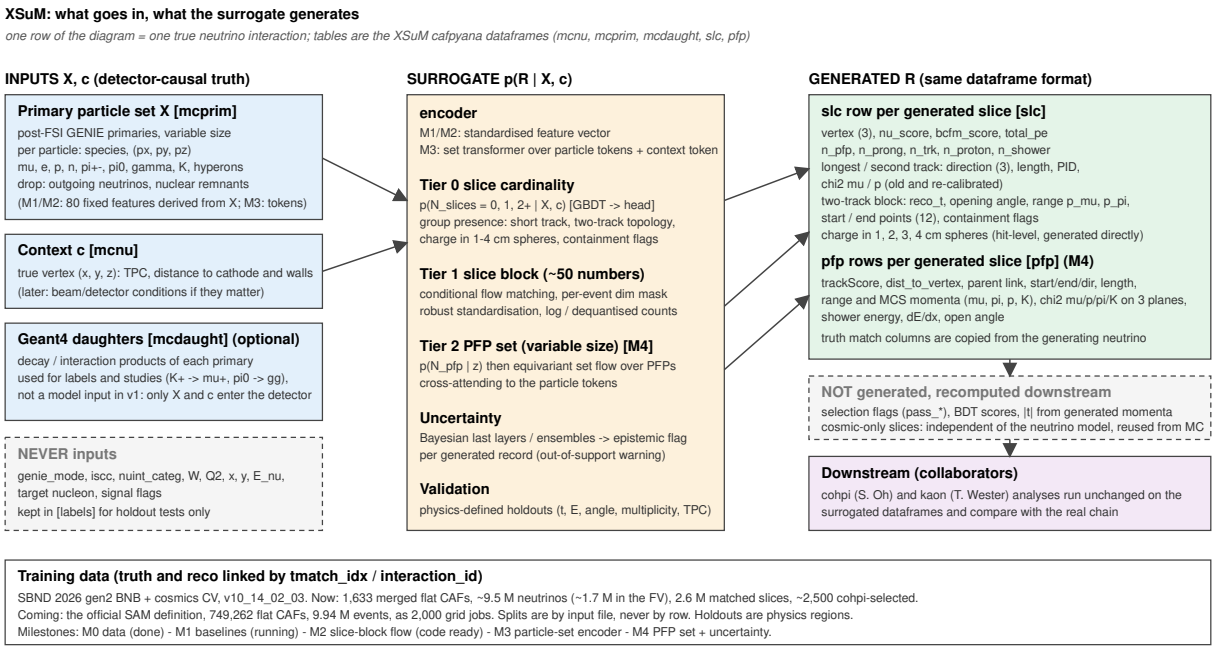


Figure 1: Inputs, surrogate, generated record and downstream use. One row of the diagram is one true neutrino interaction.

2 Data

Production. A cafpyana df-maker configuration (`analysis_village/xsum/`) writes, per flat CAF: `mcnu` (every true neutrino with the full `make_mcdf` truth, the cohpi kinematics and category), `mcprim` (every post-FSI primary), `mcdaught` (the Geant4 daughters of every primary), `slc` (every non-clear-cosmic slice with the cohpi variables and each selection stage as a flag), `pfp` (every PFP of those slices with track, shower and PID variables, kaon PID on all planes, parent links and truth match), `hdr` and POT tables. Two productions:

- **scratch1633**: the 1,633 merged flat CAFs (2.5 GB, about 3,300 triggers each) still present on the dCache scratch area referenced by the cafpyana sample list (1,363 of the 2,996 had

been purged), run in pool mode on `sbndbuild03` with 30 workers in about 3.5 h: 15.5 GB of dataframes, 6,000,000 neutrinos in the five chunks used here. This production predates the `mcdaught` table, the kaon PID branches and the removal of the 50 cm vertex-distance cut on PFPs.

- **official**: the SAM definition `aurora_SBND2026A_gen2_BNBLight_prodgenie_corsika_proton_rockbox0p1_sl` (749,262 flat CAFs, 9.94 M triggers, 7.9 TB), submitted as 2,000 grid jobs on 2026-10-16 after a 375-file pilot (21.5 s per file single-threaded, 4,500 CPU-hours in total).

A defect of the official production and its repair. The module version used on the grid counted showers among all PFPs of a slice instead of those within 50 cm of the vertex, so the two-track topology never fired and the slice two-track block and the selection flags are empty in the archive. All of them are functions of the PFP table, so they were rebuilt offline exactly as cafpyana computes them (`xsum.prep.repair`) and stored as sidecar `slc` files; the training tables and the surrogated dataset use the repaired tables. The module is fixed for future productions.

Truth–reco link. A slice points to its truth-matched neutrino through `tmatch_idx`; a PFP points to the true particle it was matched to and, through `interaction_id`, to its neutrino. A neutrino can have zero, one or several matched slices; the table keeps the slice with the highest match efficiency and records the count.

Splits. Train / validation / test are 80 / 10 / 10 by input file, never by row. In the five scratch chunks: 4,800,348 / 600,149 neutrinos in train / test. Holdout tests use physics regions (momentum transfer, energy, angle, multiplicity, TPC), see Sec. 8.

3 Inputs and targets

Inputs. The post-FSI primary particles (species, (p_x, p_y, p_z); neutrinos and nuclear remnants dropped) and the true vertex. The M1 and M2 models use 80 fixed features derived from them: the vertex and its distances to the cathode and the walls; per species (mu, e, p, n, $\mathrm{pi}^{\pm}$, pi^{0}, gamma, $\mathrm{K}^{\pm}$, K^{0}, Lambda, Sigma) the multiplicity above the cohpi thresholds (27 / 30 / 20 MeV for mu / pi / p), the summed and maximum kinetic energy and the leading particle's momentum, $\cos\theta$ and p_T; and totals (charged multiplicity, visible and total kinetic energy, electromagnetic energy, longitudinal momentum). The particle-set encoder (M3) will consume the particles directly.

Targets. Tier 0: the number of matched slices and, per slice, which groups of variables exist (a second track, the two-track topology, hits within 1–4 cm of the vertex) and the two containment flags. Tier 1: the slice block (Table **??** lists the columns). Tier 2 (M4): the PFP set.

4 M1 baselines

Gradient-boosted classifiers for the discrete targets and mixture density networks (MDN, eight diagonal Gaussians, against a single Gaussian) for continuous responses, on the 80 features. Results on the test files of the official sample (reports/m1_official); the scratch-sample results are in the git history of this note.

Slice efficiency. 32.1% of all true neutrinos (dirt and out-of-volume included) have a truth-matched slice. Log loss 0.1822 against a marginal 0.6282, AUC 0.977. The calibration against vertex z, muon momentum, visible energy and multiplicity is shown in Fig. 2.

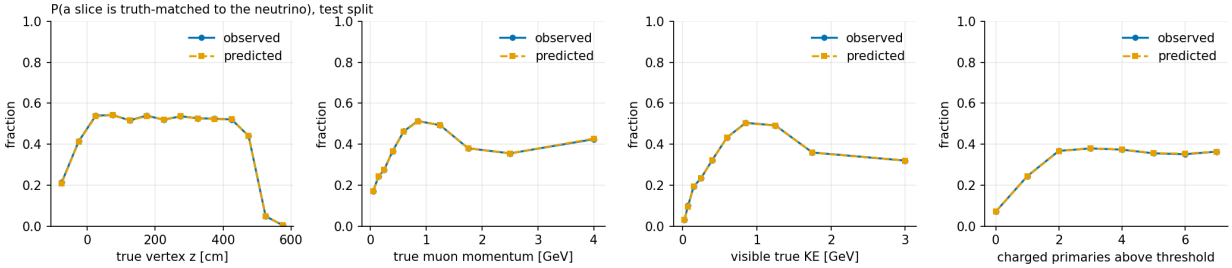


Figure 2: Observed and predicted probability that a slice is truth-matched to the neutrino, test files.

cohpi selection as a yardstick. The full cohpi selection (including the sphere-charge and cathode-crossing cuts) keeps 0.034% of neutrinos. As a check of the production, the selected count per POT agrees with the technote cut flow within 10%. The classifier reaches log loss 0.00275 against 0.00273 (AUC 0.780); per true category:

True category	test neutrinos	selected [%]	predicted [%]
out of FV	492,367	0.000	0.025
NC	30,344	0.053	0.032
signal	149	22.148	37.118
CCQE	40,886	0.024	0.022
2p2h	10,703	0.056	0.026
RES	21,548	0.427	0.321
DIS	4,152	0.554	1.026

PFP multiplicity of the matched slice: log loss 1.087 against 1.561, accuracy 55.7% against 31.8% (Fig. 3).

Continuous responses. Table 1 gives the test negative log-likelihood and Table 2 the marginals; Figs. 4–6 show them.

Target	training rows	Gaussian NLL	MDN ($K = 8$) NLL
vertex residual (3-d)	727,641	7.631	1.264
leading track: $\Delta\cos\theta$, log length	709,214	0.054	-2.501
two-track: log range-p ratios, $\Delta\cos\theta_\mu$	13,435	0.744	-1.608

Table 1: M1 MDN test NLL per target.

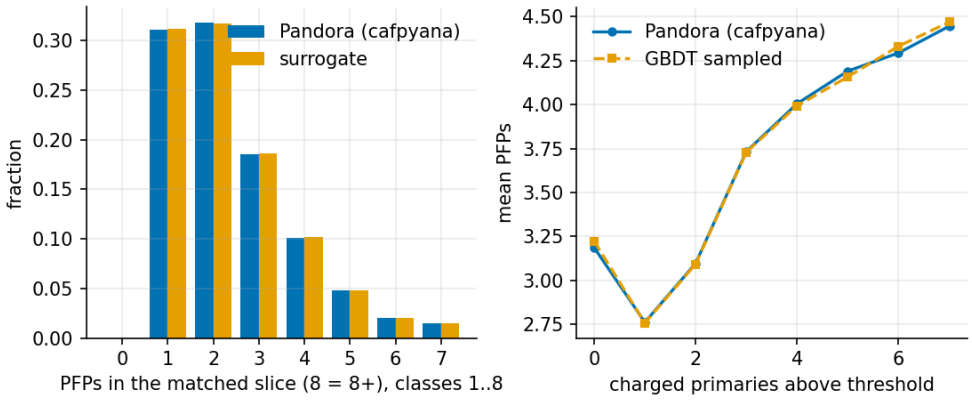


Figure 3: PFP multiplicity of the matched slice: marginal and mean versus charged primaries.

Variable	Pandora median [16%, 84%]	MDN median [16%, 84%]	W_1
dvtx_x [cm]	0.001 [-0.202, 0.213]	-0.001 [-0.205, 0.210]	2.3311
dvtx_y [cm]	0.002 [-0.378, 0.400]	0.001 [-0.364, 0.388]	3.0223
dvtx_z [cm]	-0.012 [-0.397, 0.488]	-0.010 [-0.376, 0.463]	3.8706
long_dirz - lead cos theta	-0.001 [-0.048, 0.061]	-0.002 [-0.050, 0.059]	0.0636
log(long track length [cm])	4.730 [3.674, 5.494]	4.728 [3.655, 5.501]	0.0086
log(range_p_mu / p_mu_true)	-0.015 [-0.690, 0.082]	-0.016 [-0.636, 0.148]	0.0355
log(range_p_cpi / p_pi_true)	-0.488 [-1.098, -0.132]	-0.514 [-1.074, -0.112]	0.0428
long_dirz - cos theta_mu	-0.000 [-0.030, 0.059]	0.000 [-0.033, 0.060]	0.0303

Table 2: Marginals of the MDN samples against Pandora, test files.

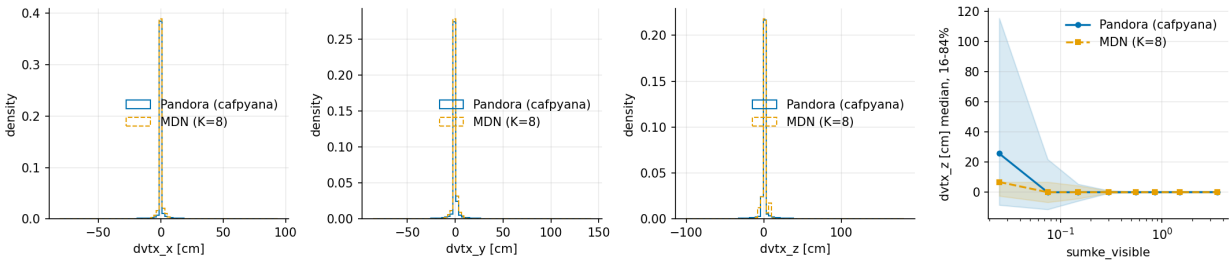


Figure 4: Vertex residual of the matched slice (FV neutrinos).

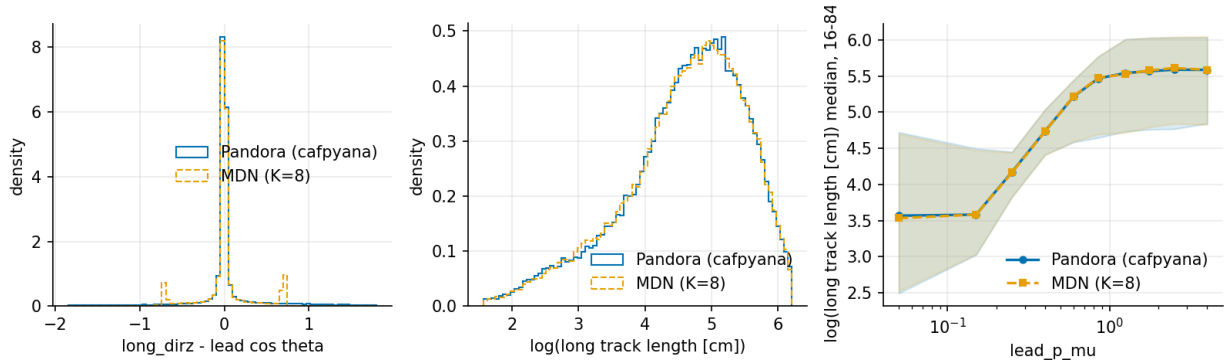


Figure 5: Longest track: direction relative to the leading charged primary and length.

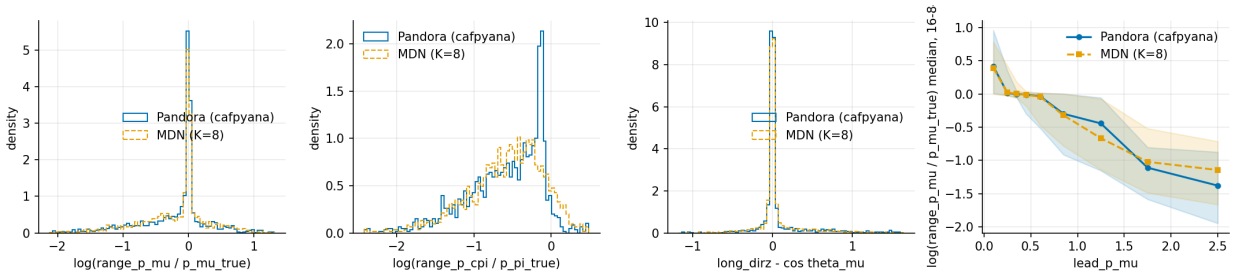


Figure 6: cohpi two-track response on 1mu1pi events passing the two-track topology.

5 M2: flow-matching surrogate of the slice block

Generation proceeds in three steps. (i) A *topology* head: a softmax classifier over the joint pattern of PFP counts (`n_pfp, n_prong, n_trk, n_proton, n_shower`; the 120 most frequent tuples plus "other", whose members are drawn from the training pool), sampled first; its one-hot encoding conditions everything that follows. (ii) Gradient-boosted heads for the presence of the variable groups that may not exist in a slice (second track, two-track block, hits within 1–4 cm of the vertex), for the two containment flags, and for the point masses at detector boundaries (vertex pinned at $z = 0$ or at the x, y walls, back-to-back track pairs). (iii) A conditional flow-matching model (linear path, Gaussian source, 6-layer velocity MLP of width 1024) over the standardised continuous block, masked per event to the groups and dimensions that exist (sim2reco `dim_mask`). Positive quantities enter as logarithms, direction cosines and the opening angle through artanh (and direction vectors are renormalised), standardisation is robust (median, MAD). Inverse transforms produce a dataframe with the production column names. Trained on 5,736,854 matched slices of the official sample for 60 epochs (54 min on one RTX 3090). Host memory is kept near 10 GB by splitting the table once on disk into a matched-only table and a 4 M-row subsample for the cardinality head, training the presence heads on 2 M-row subsamples and building the topology one-hot per batch on the GPU.

Earlier versions and what they taught: v1 (unbounded cosines, counts as dequantised flow dimensions) closed the cohpi cut flow only to 45% in the signal region; v2 (bounded cosines, point-mass heads) to 74%, losing most at the exact-count stage; v3 (topology head) closes every stage within statistics on the scratch sample and to better than 10% at every stage on the official sample (Table 4).

Group / flag	rate [%]	log loss (marginal)	log loss (GBDT)	AUC
short	60.9	0.6693	0.0858	0.995
two	5.2	0.2053	0.0172	0.999
s1	86.8	0.3909	0.3298	0.776
s2	95.3	0.1907	0.1491	0.832
s3	97.2	0.1275	0.0938	0.864
s4	98.1	0.0949	0.0665	0.885
is_muon_contained	43.3	0.6842	0.4050	0.892
is_cpi_contained	80.9	0.4878	0.2879	0.908
spike_vtx_z	13.0	0.3855	0.0948	0.985
spike_vtx_x	5.5	0.2131	0.0507	0.989
spike_vtx_y	5.5	0.2117	0.0623	0.982
spike_opening_angle	2.6	0.1189	0.0875	0.872

Table 3: Group presence and containment heads, test files.

Closure on the test files. Table **??** compares every generated column with Pandora (fraction present and Wasserstein-1 in transformed units; median W_1 0.014, maximum 4.727). Table 4 recomputes the cohpi cut flow on the generated and the real test slices, for all slices and for true signal; this is the kind of comparison the analysers will make on the full surrogated sample.

Correlations. A surrogate that matched every marginal but sampled the variables independently would fail any analysis that cuts on several of them. Fig. 8 compares the Spearman correlation

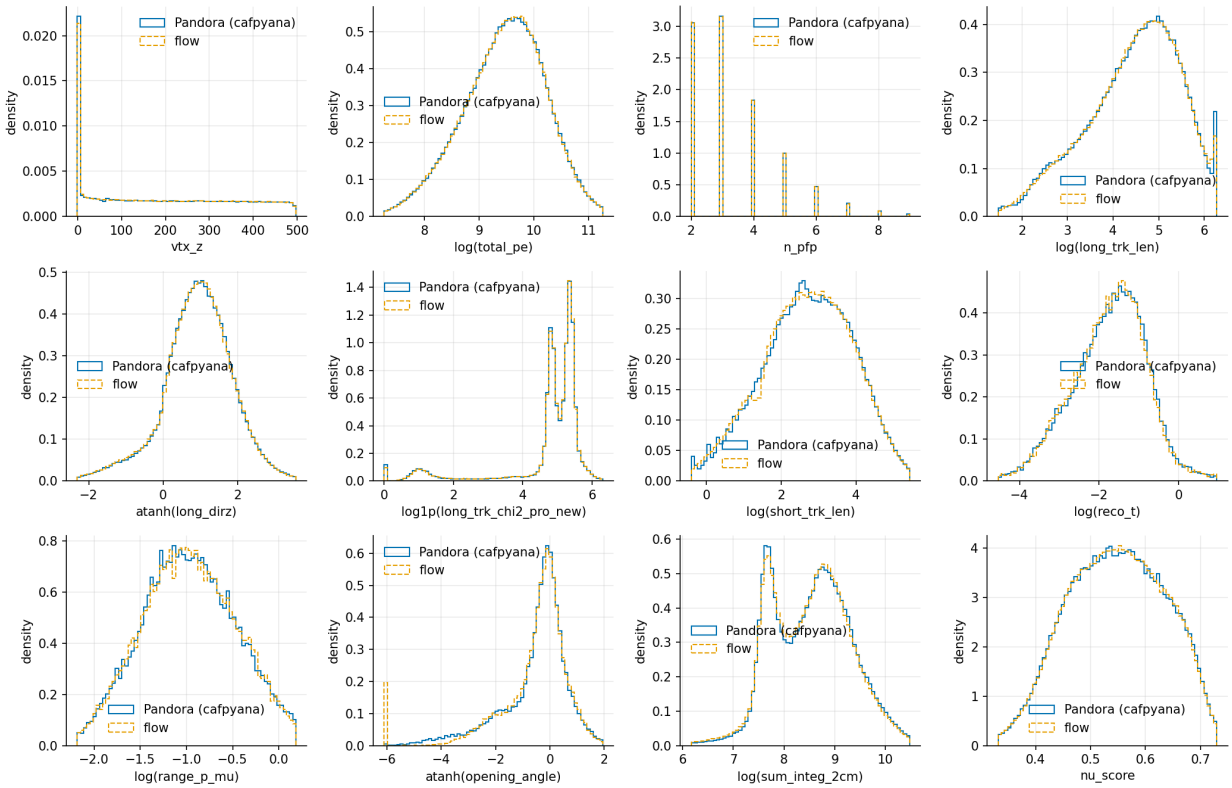


Figure 7: Marginals of generated (flow) against Pandora slice variables, test files.

matrix of 19 slice variables on 250,000 test slices: the generated matrix differs from Pandora by 0.007 on average and by at most 0.044 (all 167 defined pairs within 0.05; blank cells are pairs that are undefined because the two-track variables only exist when `n_trk`= 2). Fig. 9 shows joint densities of reco–reco pairs, including the near-deterministic range–length relation, the two-population structure of the vertex charges and of the proton chi2, and the —t——opening-angle anticorrelation that defines the cohpi signal region. Fig. 10 shows the conditional the surrogate exists to learn, reconstructed against true quantities on the same neutrinos: the exiting-muon band below the diagonal in range momentum, the mirrored band of muon/pion mis-assignment in the direction, the vertex pile-up at $z = 0$ for out-of-volume interactions, and the two populations of vertex charge.

Stage	all, Pandora	all, generated	signal, Pandora	signal, generated
pe_bcfm	453,170	454,384	483	479
2trk	38,335	38,211	226	232
dir	2,169	2,229	186	194
chi2	1,333	1,332	177	174
sphere	770	714	163	157
cathode	638	590	135	139
sr	310	289	124	119

Table 4: cohpi cut flow recomputed on real and generated test slices.

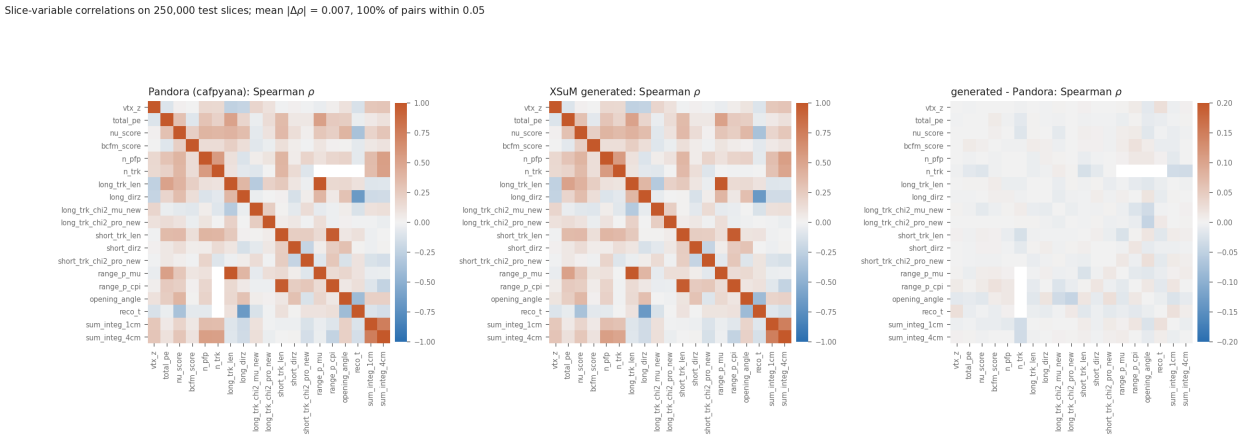


Figure 8: Spearman correlation matrices of slice variables, Pandora (left), XSuM (middle), difference (right, scale ±0.2), test slices.

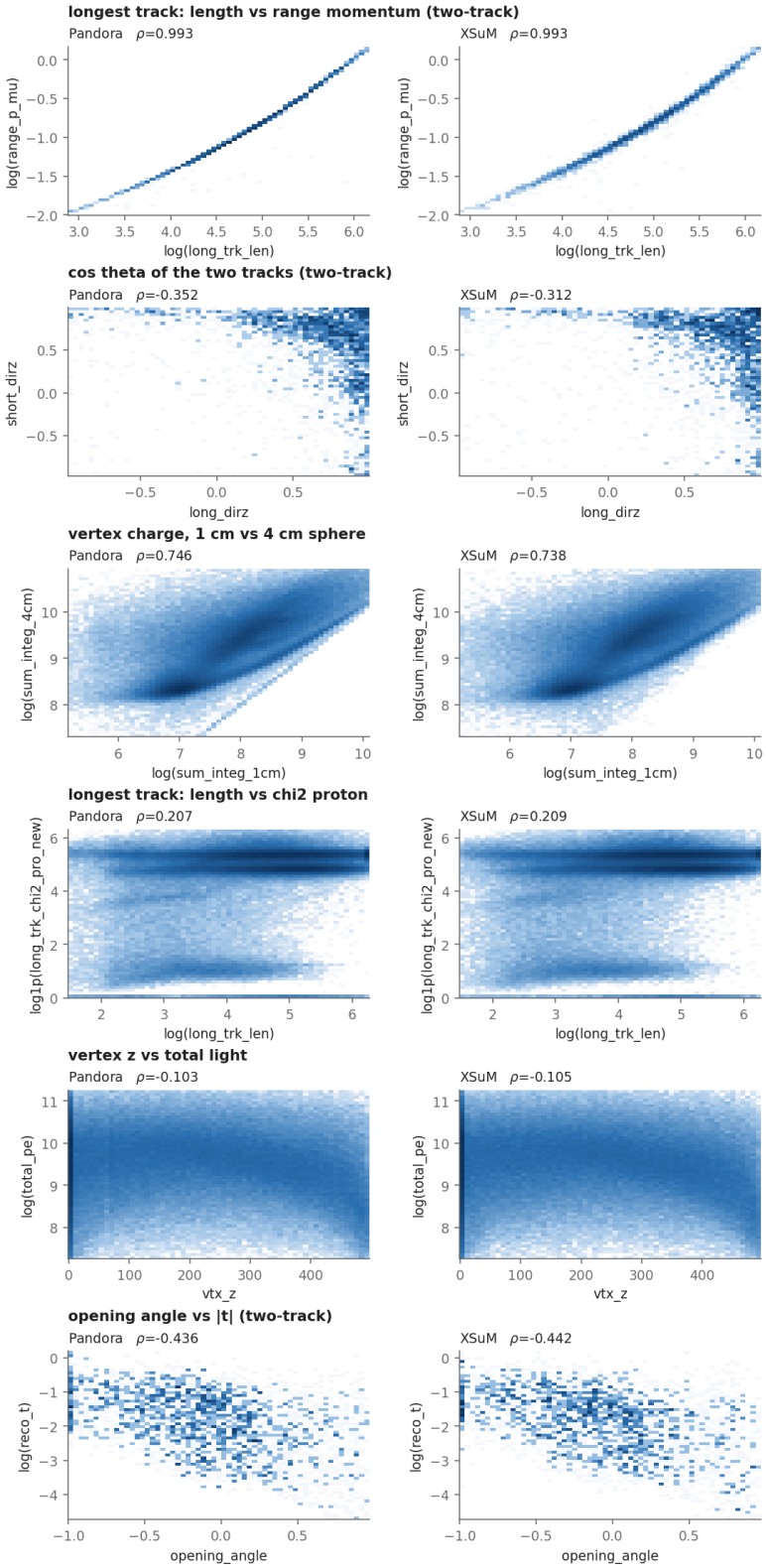


Figure 9: Joint densities of pairs of reconstructed variables, Pandora (left) and XSuM (right), same test neutrinos, same bins and color scale.

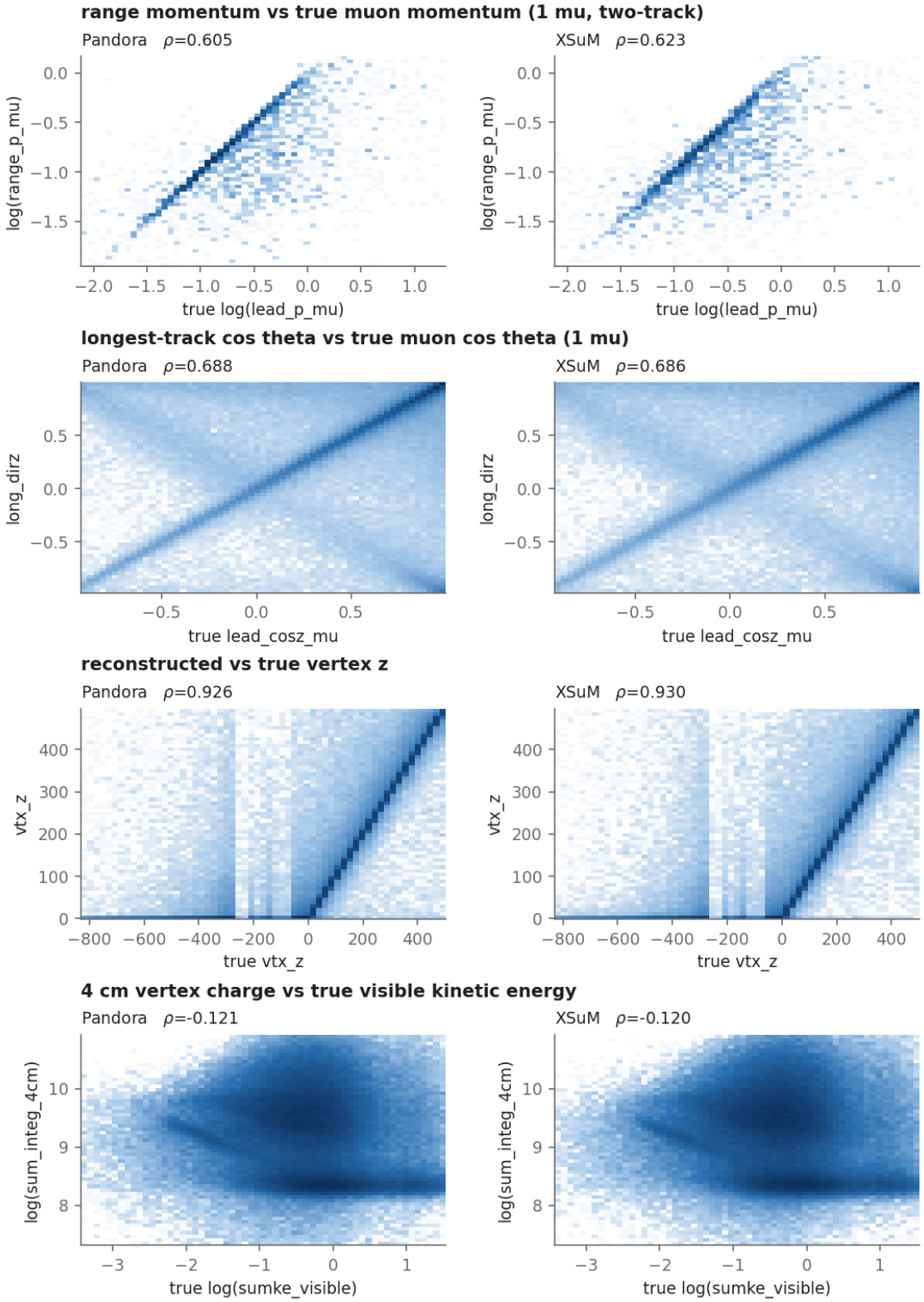


Figure 10: Reconstructed against true quantities, Pandora (left) and XSuM (right), same test neutrinos.

Column	present, real [%]	present, generated [%]	W_1 (transformed units)
n_pfp	100.0	100.0	0.0032
n_prong	100.0	100.0	0.0007
n_trk	100.0	100.0	0.0012
n_proton	100.0	100.0	0.0008
n_shower	100.0	100.0	0.0006
vtx_x	100.0	100.0	0.4322
vtx_y	100.0	100.0	0.3711
vtx_z	100.0	100.0	0.2498
nu_score	100.0	100.0	0.0005
bcfm_score	99.3	100.0	0.0017
total_pe	99.6	100.0	0.0019
long_dirx	99.9	100.0	0.0020
long_diry	99.9	100.0	0.0054
long_dirz	99.9	100.0	0.0078
long_trk_len	99.9	100.0	0.0036
long_trk_pid	99.9	100.0	0.0007
long_trk_chi2_mu	99.9	100.0	0.0100
long_trk_chi2_pro	99.9	100.0	0.0147
long_trk_chi2_mu_new	99.9	100.0	0.0099
long_trk_chi2_pro_new	99.9	100.0	0.0143
short_dirx	60.9	60.9	0.0039
short_diry	60.9	60.9	0.0067
short_dirz	60.9	60.9	0.0072
short_trk_len	60.9	60.9	0.0123
short_trk_pid	60.9	60.9	0.0026
short_trk_chi2_mu	60.9	60.9	0.0145
short_trk_chi2_pro	60.9	60.9	0.0237
short_trk_chi2_mu_new	60.9	60.9	0.0137
short_trk_chi2_pro_new	60.9	60.9	0.0225
reco_t	5.2	5.1	0.0215
opening_angle	5.2	5.1	0.0642
range_p_mu	5.2	5.1	0.0049
range_p_cpi	5.2	5.1	0.0079
muon_start_x	5.2	5.1	1.4483
muon_start_y	5.2	5.1	0.8414
muon_start_z	5.2	5.1	2.0041
muon_end_x	5.2	5.1	2.4164
muon_end_y	5.2	5.1	2.4149
muon_end_z	5.2	5.1	4.7265
cpi_start_x	5.2	5.1	1.4092
cpi_start_y	5.2	5.1	0.7554
cpi_start_z	5.2	5.1	2.0409
cpi_end_x	5.2	5.1	1.3756
cpi_end_y	5.2	5.1	0.9806
cpi_end_z	5.2	5.1	2.2161
sum_integ_1cm	86.8	86.8	0.0144
sum_integ_2cm	95.3	95.3	0.0211
sum_integ_3cm	97.2	97.2	0.0156
sum_integ_4cm	98.1	98.1	0.0203

Generation. `scripts/generate_slc.py` loads the trained surrogate and writes slice records for any truth table: slice cardinality (0, 1 or 2 per neutrino) from its head, then topology, presence, point masses and the flow, then the production column names, plus the recomputed cohpi flags. On the 814,028 test neutrinos of the five chunks it generates 269,841 slices in 66 s on one GPU (about 12,000 neutrinos per second, so the full official sample takes a quarter of an hour), with the slice-multiplicity fractions (0 / 1 / 2 slices: 68.0 / 31.0 / 1.1%) reproduced to 0.1% and 422 slices passing the full cohpi selection against 423 for Pandora (216 against 220 in the signal region).

6 M4: the PFP set of a slice

The particle-flow objects of a slice are generated as a set. Per matched slice the fitted PFPs (those with a track or shower fit; the few without either are counted separately) are sorted by distance to the vertex and padded to 12 (99.99% of slices have at most 12). Each PFP carries 29 fields: presence flags (track fit, shower fit, MCS fit, primary), distance to the vertex, track score, start point, direction, track length, range and MCS muon momenta, chi2 for muon, proton (both calibrations), kaon and pion on the collection plane, PIDA and the chi2 degrees of freedom, and the shower energy, length, opening angle, density, conversion gap and dE/dx. Flags are dequantised, fields that do not exist for a PFP are filled with noise and restored from the flags after generation (sim2reco's prong convention). The conditioning vector (245 numbers) is the truth features, the topology one-hot and the generated slice block, so the PFP set is consistent with the slice record. Two softmax heads give the number of fitted and of empty PFPs; a masked set flow (transformer velocity net over the PFP tokens, flow matching) generates the records. Trained on 3,000,000 slices for 40 epochs.

The number of fitted PFPs is predicted with 93.1% accuracy (log loss 0.177 against 1.552 marginal): the slice block already carries the multiplicities. Table **??** compares every generated field with Pandora on the test slices (median W_1 0.005, maximum 0.830, in transformed units), Fig. 11 shows marginals. The physics closure is Table 6: the cohpi multiplicities (prongs, track candidates, proton candidates, showers) recomputed from the generated PFPs against the same recomputation on the real PFPs; the exactly-two-tracks topology rate is 8.03% generated against 7.97% real (7.97% from the slice table itself).

Count	mean, Pandora	mean, generated	0/1/2/3/4, Pandora	0/1/2/3/4, generated
n_prong	1.587	1.589	4344 / 161854 / 97884 / 28326 / 6499	4404 / 161669 / 97979 / 28087
n_trk	1.439	1.442	18228 / 165064 / 90105 / 22177 / 4166	18486 / 164604 / 89919 / 22271
n_proton	0.409	0.408	194597 / 90844 / 13320 / 1559 / 159	194894 / 90586 / 13100 / 1742
n_shower	0.374	0.377	222945 / 51455 / 19369 / 5085 / 1262	223694 / 49801 / 19592 / 5631

Table 6: cohpi multiplicities recomputed from PFP sets, test slices.

Field	Pandora median (flags: mean)	generated	W_1 (transformed units)
has_trk	1.000	1.000	–
has_shw	0.999	1.000	–
has_mcs	0.707	0.708	–
is_primary	0.927	0.932	–
dist_to_vertex	0.549	0.553	0.0123
trackScore	0.191	0.191	0.0020
start_x	-18.933	-18.711	0.7015

Field	Pandora median (flags: mean)	generated	W_1 (transformed units)
start_y	1.692	1.873	0.7898
start_z	221.122	221.582	0.8300
dir_x	0.008	0.010	0.0012
dir_y	0.005	0.006	0.0020
dir_z	0.604	0.605	0.0037
trk_len	3.292	3.286	0.0064
rangeP_mu	-1.792	-1.794	0.0027
mcsP_mu	-0.916	-0.929	0.0145
chi2_mu	2.628	2.626	0.0045
chi2_pro	4.894	4.894	0.0027
chi2_mu_new	2.629	2.625	0.0049
chi2_pro_new	4.906	4.904	0.0035
chi2_kaon	3.924	3.929	0.0044
chi2_pion	2.954	2.952	0.0046
pida	2.032	2.031	0.0052
chi2_ndof	3.332	3.301	0.0231
shw_energy	-2.234	-2.238	0.0043
shw_len	3.128	3.132	0.0051
shw_open_angle	0.026	0.026	0.0039
shw_density	1.593	1.595	0.0029
shw_conversion_gap	0.539	0.544	0.0122
shw_dEdx	0.973	0.984	0.0136

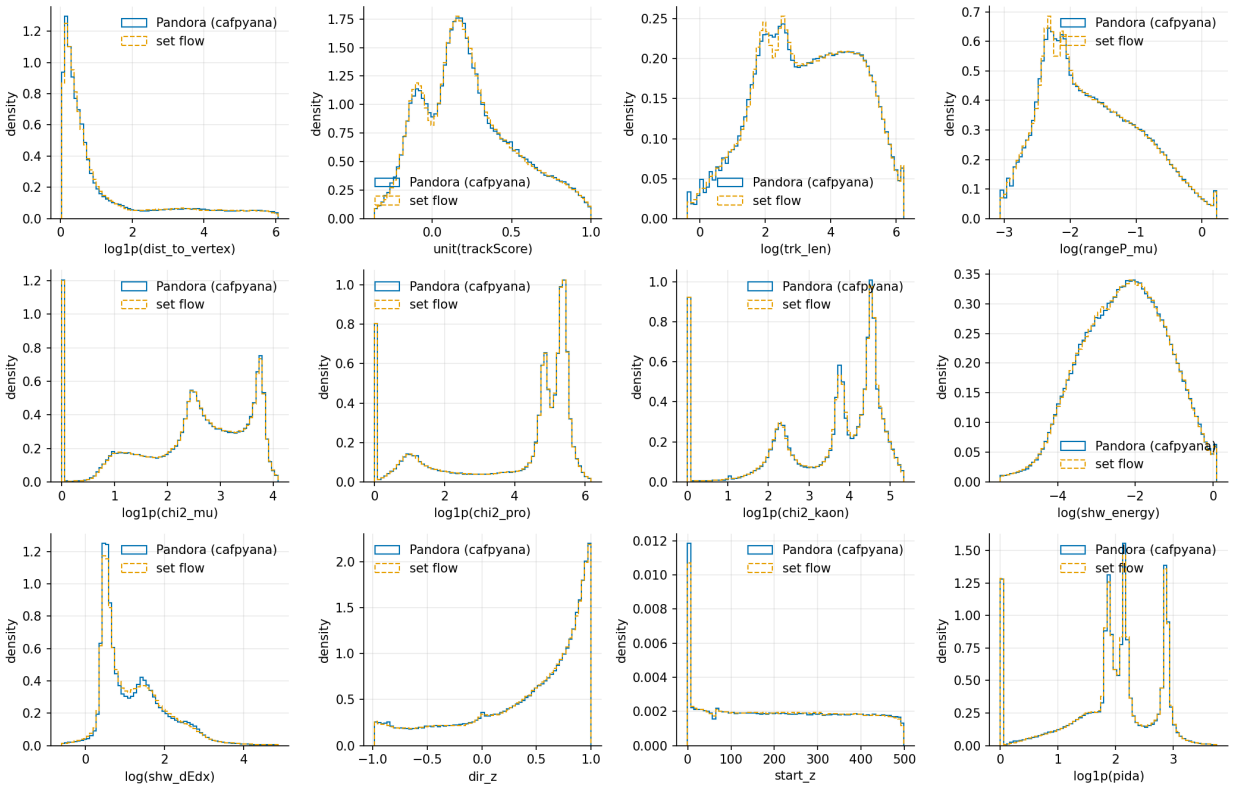


Figure 11: PFP fields, generated set against Pandora, test slices (valid tokens).

7 Surrogated datasets for the analysers

`scripts/write_surrogate_df.py` turns the generated slices into files in the production format, so that existing cafpyana-based analysis code runs on them unchanged. The first dataset, `surrogate_v3_6ch`, applies the six-chunk M2 v3 model to the 9,774,979 true neutrinos of the scratch production (the training sample itself; comparisons with the real chain must use the test files listed in `file_splits.parquet`, 10% of the input files chosen by file):

- location: `/pnfs/sbnd/persistent/users/lwan/dataframe/surrogate_v3_6ch/`, six files `xsum_surrogate_c` (11 GB), world-readable; the real production it mirrors is in `prod_v10_14_02_03_scratch1633/` (18 GB) next to it;
- keys per file: `mcnu_0, mcprim_0, hdr_0, histpotdf_0, histgenevtdf_0, pot_0` copied from the production, `slc_0` generated (index `(__ntuple, entry, rec.slc..index)`, `tmatch_idx` = generating neutrino, `tmatch_eff` / `tmatch_purity` NaN, the cohpi `pass_*` and `in_sr` flags recomputed from the generated variables, plus `surrogate_topology_class`), `provenance`;
- content: 3,241,308 generated slices (13 min of generation), of which 5,252 pass the full cohpi selection and 2,759 are in the signal region; no `pfp` table yet (M4), no cosmic-only slices (to be taken from the MC).

The second dataset, `surrogate_official_v3` (2026-10-22), applies the official-sample model of Sec. 5 to all 17,861,754 neutrinos of the official production: 5.9 M generated slices in 27 min, 6,302 of which pass the full cohpi selection against 6,095 in the real chain (2,988 in the signal region). It is

written as 2,000 files `xsum_surrogate_N.df` (75 GB) in `/pnfs/sbnd/persistent/users/lwan/dataframe/surroga` one per grid job of the archive `prod_v10_14_02_03_official/`, with `mcnu, mcprim, mcdaught, pfp, hdr` and the POT tables copied from the production and `slc_0` generated (same conventions as above); `file_splits.parquet` marks the 74,533 test files. Note that the real `slc` tables of that archive must be read from the repaired sidecars `prod_v10_14_02_03_official_slcfix/` (Sec. 2).

`surrogate_official_v3_pfp` (2026-10-22) is the same dataset with a generated `pfp_0` table from the M4 v2 set model of Sec. 6: 14,005,625 PFPs for the 5,926,843 generated slices (69 min of generation), with the 26 generated fields laid out under the production column names, straight-line track end points, no PFPs without a fit, and no induction-plane chi2 or truth-match columns. This is the first dataset in which every reconstructed object of the slice is surrogated.

8 Outlook

1. Official sample (9.94 M triggers): retrain M1/M2, add the `mcdaught` and full-PFP tables.
2. M3: particle-set encoder (transformer over primaries + vertex token) replacing the fixed features.
3. M4 (first version in Sec. 6): generated `pfp` tables in the surrogated datasets, chi2 on the induction planes and track end points, cross-attention to particle tokens (M3), Bayesian last layers.
4. Holdouts: train on $|t| < 0.1$, test the tail; low versus high E_ν; forward versus wide-angle muons; ≤ 2 versus ≥ 3 charged primaries; one TPC versus the other.
5. Surrogated dataframes written in the production format for the cohpi and kaon analysers.